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Key Personnel Changes/Previous PI:			
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Task Description:	Visual Impairment /Intracranial Pressure (VIIP) [Ed. note: now known as Spaceflight Associated Neuro-ocular Syndrome (SANS)] is a top human spaceflight risk for which NASA does not currently have a proven mitigation strategy. Thigh cuffs (Braslets) and lower body negative pressure (LBNP; Chibis) devices have been or are currently being evaluated as a means to reduce VIIP signs and symptoms, but these methods alone may not provide sufficient relief of cephalic venous congestion and VIIP symptoms. Additionally, current LBNP devices are too large and cumbersome for their systematic use as a countermeasure. Therefore, a novel approach is needed that is easy to implement and provides specific relief of symptoms. This investigation proposes the evaluation of an impedance threshold device (ITD) as a VIIP countermeasure. The ITD works by providing up to 7 cm H2O (~5 mmHg) resistance to inspiratory air flow, effectively turning the thorax into a vacuum pump upon each inhalation which lowers the intrathoracic pressure (ITP) and facilitates venous return to the heart. The ITD is Food & Drug Administration (FDA)-approved and was developed to augment venous return to the central circulation and increase cardiac output in cardiopulmonary resuscitation (CPR) and in patients with hypotension. While the effect of ITD on CPR survival outcomes is controversial, its ability to lower ITP with a concomitant decrease in intracranial pressure (ICP) is well documented. A similar concept that creates negative ITP during exhalation (intrathoracic pressure regulator; ITPR) decreased ICP in 16 of 20 patients with elevated ICP in a hospital pilot study. ITP and central venous pressure (CVP) have been shown to decrease in microgravity; however, ITP drops more than CVP, indicating an increased transmural CVP. This could explain the paradoxical distention of jugular veins (JV) in microgravity despite lower absolute CVP and also suggests that JV transmural pressure is not dramatically elevated; JV pressure may be lowered sufficiently to remove the cephalic venous congestion with an ITD. During spaceflight experiments with Braslet thigh cuffs and modified (open-glottis) Mueller maneuvers, Braslet reduced cardiac preload but only reduced the internal JV (IJV) cross sectional area by 23%. Addition of Mueller maneuvers resulted in an IJV area reduction of 48%. The concept presented in this proposal is that application of an ITD essentially applies a Mueller maneuver with added negative ITP in every respiratory cycle, acting to: 1) reduce venous congestion in the neck and 2) lower ICP. The expected mechanism of action is that in microgravity (or an analog) blood is relocated toward the heart from the head and neck. Once validated, the ITD would be an exceptionally easy countermeasure to deploy and test on the International Space Station (ISS). Dosage could be easily altered through 1) duration of application and 2) inspiratory resistance set point. Effects could be additionally enhanced through co-application with other countermeasures such as thigh cuffs. The experimental approach is to use a battery of tests that are currently being used to evaluate effects of other interventions. Healthy test subjects will be asked to participate in two sessions, one with an ITD and one with a sham ITD (placebo). Subjects will be evaluated in the seated and supine positions as well as head down tilt (HDT) postures, including 6 and 15 degree HDT. Measures taken at each posture will include IJV cross sectional area, IJV and carotid artery Doppler, transcranial Doppler (TCD), Optic nerve sheath diameter (ONSD), Superior Ophthalmic Vein (SOV) Doppler, optical coherence tomography (OCT), cochlear and cerebral fluid pressure analysis (CCFP), intraocular pressure (IOP), facial soft tissue thickness and hyperemia, arterial blood pressure, and heart rate.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has established the ITD as a viable means to reduce venous congestion while concomitantly restoring normal blood flow patterns in cephalic circulation. SANS prevention and therapy development, as for any other medical condition, requires understanding of its etiology and pathogenesis. The stark diversity of SANS expression among long-duration crew, in terms of both manifestations and severity, has not been explained. We conducted a blinded placebo-controlled trial of a physical intervention that acts upon one of the most obvious and well-established phenomenon of microgravity physiology – venous stasis in the brachiocephalic region, which appears to be a primary contributor to SANS development. We conclude that 1) ITD breathing is effective for venous decongestion; 2) ITD changes ocular blood flow towards seated values, and 3) ITD is safe and well-tolerated by subjects. This acute ground-based study did not demonstrate reduction of non-invasive ICP indicators with ITD, nor did it show a significant difference in IOP or choroid engorgement between ITD intervention and placebo in HDT (however, recent publications have confirmed that ICP is lowered with ITD breathing). The results of this study warrant continued attention to inspiratory resistance as a promising intervention for SANS therapy and possibly, prevention of its development. Furthermore, abnormal flow patterns in IJV that we observed during exaggerated venous stasis of extreme HDT, deserve thorough, focused examination.
Task Progress:	INTRODUCTION: Spaceflight associated neuro-ocular syndrome (SANS) is a top human spaceflight risk, for which NASA does not yet have a proven mitigation strategy. Thigh cuffs (Braslet) and lower-body negative pressure (LBNP; Chibis) devices have been evaluated as a means to reduce SANS signs and symptoms, but these methods alone may not provide sufficient relief of cephalic venous congestion and prevent SANS symptoms. Additionally, current LBNP devices are large and do not allow crewmembers to move freely about the cabin while in use, limiting their ability to be used routinely as a countermeasure. Therefore, a novel countermeasure is needed that is easy to implement and provides specific relief of symptoms. This investigation evaluated the impedance threshold device (ITD) as a SANS countermeasure. The ITD provides up to 7 cm H2O (~5 mm Hg) resistance to inspiratory airflow, lowering the intrathoracic pressure (ITP) and effectively augmenting the function of thorax as a vacuum pump during inspiration. The ITD was developed to augment venous return to the central circulation and increase cardiac output during cardiopulmonary resuscitation (CPR) or in patients with hypotension, and is approved by the United States Food and Drug Administration (FDA) for that purpose. While the effect of an ITD on CPR survival outcomes is controversial, the ITD's ability to lower ITP with a concomitant decrease in intracranial pressure (ICP) is well documented. We expect that use of the ITD in weightlessness (or an analog) will counter the cephalad fluid shift by moving blood from head and neck vasculature towards the heart. We hypothesized that ITD breathing (similar to repeated Mueller maneuvers) in supine or head-down subjects would reduce ITP during the respiratory cycle, serving to reduce venous congestion in the neck and lower ICP. METHODS: Fifteen healthy test subjects each participated in two trials, one with an ITD and one with a sham ITD (placebo), in multiple postures: seated, supine, 6° head-down tilt (HDT), 15° HDT, and extended 15° HDT (additional 30 minutes). Measures analyzed from each posture included internal jugular vein (IJV) cross-sectional area (CSA), noninvasive IJV pressure (VeinPress), superior ophthalmic vein direction of flow, choroid thickness (using Optical Coherence Tomography (OCT)), central retinal artery (CRA) and vein (CRV) Doppler, noninvasive estimates of ICP (cochlear and cerebral fluid pressure analyzer and otoacoustic emissions), intraocular pressure (IOP), arterial blood pressure, and heart rate. Questionnaires were administered at each posture to document subjective status.

Task Progress:	RESULTS AND DISCUSSION: All sham postural profiles demonstrated expected trends (e.g., parameters such as IJV CSA, noninvasive IJV pressure, and ICP estimates all increased with increasing HDT). IJV CSA and JV pressures were decreased with ITD in most postures, supporting the hypothesis that ITD facilitates drainage of the head and neck. CRA and CRV velocities increased with increasing HDT, but were decreased with ITD. The most logical explanation for this response is the reduction in vessel diameter during HDT, with a re-expansion induced by ITD. Constraint of the CRA and CRV is mediated by the optic nerve and sheath, however is not necessarily due to ICP alone; autoregulation and pressures in the intraconal space likely play a significant role. Choroid thickness, IOP, and noninvasive ICP estimates did not show systematic difference between ITD and sham. Lack of ICP reduction with ITD was unexpected given ample documentation of this effect in the literature. It is possible that our ear-based non-invasive ICP methods were particularly sensitive to the altered airway pressure induced by ITD, masking any changes in ICP. In similar ground-based experiments (Fluid Shifts), choroid thickness was also not reduced with LBNP. In the context of the posture-based model that we employed, our conclusions are: 1) based on IJV parameters, ITD unequivocally relieves congestion in (filling of) the large neck veins; 2) based on ocular Doppler data, ITD has favorable peripheral vascular effects in the orbit (i.e., in the direction opposite to those demonstrated in simulated or actual microgravity), and 3) our data do not demonstrate a clear reduction in ICP with ITD. From our experiences in the laboratory, the ITD or a less obtrusive similar device would be easy to deploy and test as an inflight countermeasure. Investigating several combinations of dosing (frequency and duration of use) with the level of the inspiratory resistance may lead to higher efficacy. ITD was well tolerated by subjects in the laboratory; however, future work with this device should also include addressing comfort during prolonged wear to assure compliance and acceptance. Planned forward work includes investigation of the ITD in combination with thigh cuffs and LBNP countermeasures.
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